

Question	Answer	Marks
2(a)	force $\times$ <u>perpendicular</u> distance (of line of action of force to / from the point)	B1
2(b)(i)	(moment due to weight $=$) $1.2 \times 270 \times 9.81$	B1
	(moment due to post $=$) $1800 \times (1.6 / \cos \theta)$	B1
	$1.2 \times 270 \times 9.81 = 1800 \times (1.6 / \cos \theta)$ so $\theta = 25(^{\circ})$ or $1.2 \times 270 \times 9.81 - 1800 \times (1.6 / \cos \theta) = 0$ so $\theta = 25(^{\circ})$	B1
2(b)(ii)	A closed tip-to-tail vector triangle	M1
	vector labelled F at an angle of $25^{\circ} \pm 3^{\circ}$ anticlockwise from vertical and vector labelled R at an angle of $37^{\circ} \pm 3^{\circ}$ clockwise from vertical	A1

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Question	Answer	Marks
2(b)(iii)	$A = F / p$	C1
	$= 1800 / (150 \times 10^3)$	
	$= 0.012 \text{ m}^2$	A1

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